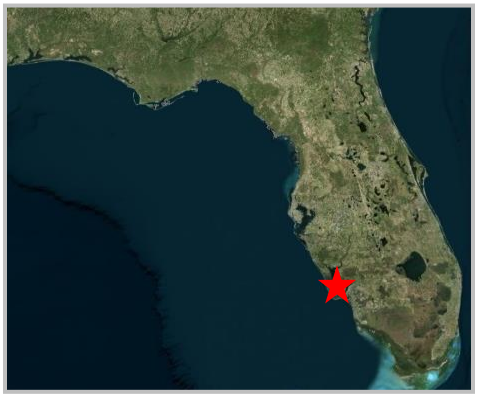
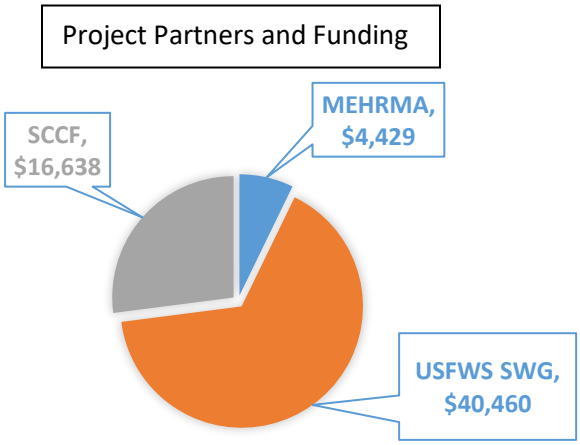
Restoring Mangrove Islands in Pine Island Sound, Lee County, FL

FY 20/21





Project Description



Figure 1: Volunteer oyster reef enhancement at Hemp Key Critical Wildlife Area.

This project restored mangrove and oyster reef habitat for fish and wildlife Species of Greatest Conservation Need in Pine Island Sound using methods demonstrated to work effectively in the area. Pine Island Sound is characterized by extensive mangrove shorelines and mangrove islands, fringing oyster reefs, and shallow seagrass beds which serve as important feeding, breeding, and nursery habitats for fish and wildlife. Repeated hurricanes and sea level rise damaged and degraded habitats on and surrounding five islands managed by the U. S. Fish and Wildlife Service (USFWS) as part of the Pine Island National Wildlife Refuge: Hemp Key, Bird Key, Benedict Key, Rat Key, and Cat Key. Restoration of resiliency and habitat function at these five islands is being achieved through planting of mangrove propagules, enhancement of oyster reefs, and structural and functional monitoring during this three-year project. Hemp Key is designated by FWC as a Critical Wildlife Area because it supports a high density of nesting wading and diving birds.

Pilot mangrove planting, oyster reef restoration, and initial monitoring were completed by Sanibel-Captiva Conservation Foundation (SCCF) staff and volunteers supported by a State Wildlife Grant (SWG) through USFWS. AHCR provided appropriation funding that was used deliver and stage 50 cubic yards of large fossil shell that was used in oyster reef enhancement. During subsequent grant years SCCF will continue mangrove and oyster restoration and monitoring.

Benefits

- Provide foraging and nesting habitat for a variety of wildlife including wading birds, diving birds, manatee, marine turtles, fish, and crabs.

MESS Strategic Plan Elements Supported

• Strategic Action 1.1.1: Identify and develop restoration projects in marine and estuarine habitats (e.g. mangroves, seagrass, saltmarsh, oyster reef, coral reef) on publicly owned state lands including those on conservation lands acquired through non-regulatory means.
• Strategic Action 1.1.4: Implement priority restoration plans.

Acreage restored by habitat type (footprint)

Total- 4.0 acres estuarine habitat:

- Mangrove habitat- 3.0 acres
- Oyster reef habitat- 1.0 acre

Acreage affected by restoration activities: 100,388 acres of intertidal and soft-bottom estuarine habitat within the hydrologic unit.



Figure 2: Mangrove propagule planting to stabilize island shorelines and provide future wading bird nesting habitat.



Figure 3: Oyster reef functional monitoring by Sanibel - Captiva Conservation Foundation staff.